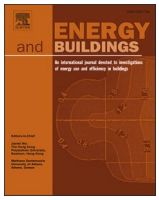
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Comprehensive effects of thermal and acoustic environments on overall comfort in urban parks of cold regions in China

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ABSTRACT

Urban parks play a crucial role in enhancing residents' quality of life; however, research on the combined effects of thermal and acoustic environments on subjective evaluation remains limited. This study explores these interactions in cold-region urban parks, focusing on People's Park in Zhangjiakou, China. A combination of field measurements and questionnaire surveys was conducted across three representative spaces characterized by three common sound types: human-generated, artificial, and natural sounds. The results indicate that, in terms of thermal perception, natural sounds significantly reduce heat stress by lowering the thermal sensation vote and improving thermal comfort, whereas artificial sounds exacerbate it. Additionally, high sound pressure levels ($L_{Aeq} \geq 60$ dB(A)) exacerbate thermal discomfort, making individuals feel hotter in summer and colder in winter, thereby reducing thermal comfort. Regarding acoustic perception, a thermally comfortable environment enhances the acoustic sensation vote and improves acoustic comfort. For overall comfort, during summer and transitional seasons, the overall comfort vote is negatively correlated with both the Universal Thermal Climate Index (UTCI) and sound pressure levels, while in winter, overall comfort vote is positively correlated with UTCI but negatively correlated with sound pressure levels. The analysis of equivalent overall comfort vote further reveals that in summer and transitional seasons, rising temperatures under extreme heat conditions (UTCI $\geq$ 35°C) have a more pronounced negative impact on overall comfort vote than decreasing temperatures in winter. Conversely, in winter, variations in sound pressure levels exert a stronger influence on overall comfort vote compared to those in summer and transitional seasons. This study addresses a research gap in understanding the interaction between thermal and acoustic environments in cold-region urban parks and provides valuable insights for the future planning and optimization of urban green spaces in such climates.

1. Introduction

Throughout human history, outdoor spaces have played an indispensable role in enhancing quality of life, improving residential comfort, and fostering social interactions [1]. High-quality outdoor environments not only provide opportunities for contact with nature and physical and mental relaxation but also help alleviate psychological stress and physical fatigue, thereby reducing the risk of related health issues [2]. However, the comfort of outdoor environments is influenced by a complex interplay of multiple environmental factors, which often interact in dynamic and sometimes contradictory ways [3]. A change in any single factor can trigger cascading effects that ultimately impact overall environmental comfort [4]. Key environmental components that influence comfort include, but are not limited to, thermal conditions [5,6], acoustic environments [7,8], lighting [9,10], and air quality [11,12]. Among these, thermal and acoustic environments have received particular attention due to their frequent and pervasive disturbances. Thermal discomfort is widely recognized as a major constraint on individuals' willingness to engage in outdoor activities [13]. Adverse thermal conditions can impair thermoregulation, induce heat stress, and significantly reduce the desire for outdoor participation, thereby negatively affecting quality of life [14]. In the context of intensifying urban heat island effects [15], improving thermal comfort in outdoor spaces has become increasingly critical. Similarly, an unfavorable acoustic environment not only causes direct harm to the auditory system but also triggers stress responses and affects emotional states, leading to long-term negative impacts on physical and mental health [16,17]. Studies have shown that sound satisfaction is a crucial component of overall satisfaction with outdoor spaces [18]. Compared to factors such as lighting and air quality, acoustic discomfort has a more immediate and pronounced negative impact on overall comfort [19]. In

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Nomenclature

UTCI	Universal Thermal Climate Index
L_{Aeq}	Equivalent continuous sound level (dB(A))
SPL	Sound pressure level
T_a	Air Temperature (℃)
RH	Relative Humidity (%)
T_g	Globe Temperature (℃)
V_a	Wind Speed (m/s)
TSV	Thermal Sensation Vote
TCV	Thermal Comfort Vote
ASV	Acoustic Sensation Vote
ACV	Acoustic Comfort Vote
OCV	Overall Comfort Vote
P1	Fitness Activity Area
P2	Ecological Recreation Area
P3	Central Waterscape Area

recent years, with rapid urbanization and the rise in traffic and social activity noise, optimizing the acoustic environment has become a critical aspect of improving outdoor space quality [20,21]. Therefore, investigating the combined effects of thermal and acoustic environments on outdoor space comfort, as well as their impact on human behavior and health, is of great significance. Such research not only contributes to increased outdoor space utilization but also provides a scientific basis for future urban planning and environmental management.

Extensive research has been conducted on the individual effects of thermal and acoustic environments in outdoor spaces, leading to the establishment of relevant building thermal environment standards, acoustic technical indices, and design guidelines [22,23]. These studies and standards provide a scientific basis for improving comfort related to specific environmental factors. However, in real-life settings, the environment is a complex system influenced by multiple interacting factors rather than isolated elements acting independently [24–28]. Despite significant progress in understanding thermal and acoustic environments separately, research on environmental evaluation under multifactor interactions remains insufficient.

In recent years, increasing attention has been given to the study of thermal-acoustic interactions, with various environmental contexts explored to understand their underlying mechanisms, as summarized in Table 1. However, a review of the existing literature reveals significant research gaps in this field. First, most studies have focused on temperate and tropical regions, while research on cold regions remains limited. This gap hinders a comprehensive understanding of the unique characteristics of thermal-acoustic interactions under extreme climatic conditions. Second, previous studies have predominantly examined urban functional areas such as plazas and campuses, whereas limited attention has been given to urban parks in cold regions, which serve as essential spaces for both ecological conservation and recreational activities. Furthermore, existing research primarily investigates the independent effects of thermal and acoustic environments on comfort, with relatively superficial discussions on the interaction mechanisms between these factors. Moreover, existing studies have predominantly focused on the independent effects of either thermal or acoustic environments on comfort, while research on the interactive mechanisms between the two remains relatively limited. Although recent studies have begun to address thermal perceptions under dynamic environmental conditions [29], comprehensive investigations that incorporate both dynamic variability and seasonal differences across broader populations and diverse outdoor settings are still limited. Urban parks in cold regions of China exhibit unique environmental complexities due to distinct seasonal variations and frequent extreme winter weather events. The impact of thermal-acoustic environments on human behavior and psychological comfort under low-temperature and snow-covered conditions remains an underexplored yet critical research area. Additionally, current studies tend to focus on single sensory dimensions, such as thermal or acoustic comfort, while overlooking the multisensory interactions among thermal, acoustic, and visual environments. This limitation restricts the ability to identify key factors influencing overall human comfort in outdoor spaces.

Against the backdrop of China's "dual carbon" goals and the push for green urban development, urban parks in cold regions play a crucial role as public spaces, offering opportunities for relaxation, recreation, and physical activity [30]. These parks significantly contribute to improving public health and overall well-being. A deeper investigation into the thermal-acoustic interactions in urban parks in cold regions can not only bridge existing research gaps but also provide scientific insights for enhancing environmental comfort, increasing the attractiveness of outdoor spaces during winter, and supporting the sustainable development of cold-region cities [31]. Therefore, studying the interaction mechanisms of thermal-acoustic environments in urban parks in cold regions is of great theoretical significance and urgent practical value.

This study focuses on an urban park in Zhangjiakou, a representative cold-region city in China, selecting three typical open spaces with distinct soundscape characteristics to comprehensively investigate the combined effects of thermal and acoustic environments on human perception and comfort under different seasonal conditions. By integrating field measurements with questionnaire surveys, this research aims to reveal the mechanisms through which thermal-acoustic environments influence multisensory experiences in urban parks in cold regions. The specific objectives of this study are as follows: (1) To analyze the effects of various thermal and acoustic conditions in open spaces within urban parks in cold regions on thermal and acoustic comfort. This will help identify patterns of thermal and acoustic perception across different environmental conditions, providing fundamental data to support improvements in outdoor comfort in cold regions. (2) To investigate the interaction between thermal and acoustic perception in open spaces of urban parks in cold regions. This includes analyzing the dynamic relationships between thermal and acoustic comfort, as well as their mutual regulatory effects. By clarifying the synergistic or inhibitory mechanisms in complex environmental settings, this study aims to establish a theoretical foundation for multisensory-integrated environmental design. (3) To quantify the influence of thermal perception, acoustic perception, and their interactions on subjective human experiences from an overall comfort perspective. This will enable the exploration of the integrative effects of multidimensional environmental perception in cold-region urban parks and inform specific strategies for optimizing thermal-acoustic environments.

The ultimate goal of this study is to develop a scientifically grounded thermal-acoustic interaction model based on the unique climatic and soundscape characteristics of cold regions. This model will provide theoretical support and technical guidance for creating comfortable environments under combined thermal-acoustic effects in open spaces of urban parks in cold regions. By addressing existing research gaps, this study will also offer valuable insights for the planning and optimization of green spaces in cold-region cities.

2. Methods

2.1. Study area

Zhangjiakou experiences a temperate continental monsoon climate characterized by four distinct seasons: long and cold winters, dry and windy springs with frequent sandstorms, short and hot summers with concentrated precipitation, and clear, mild autumns. Over the past decade (2013–2022), the average annual temperature in Zhangjiakou was 9.2 ℃, with an average temperature of −7.8 ℃ in January and 24.2 ℃ in July. The city has an annual average precipitation of 357 mm and

Table 1
Review on Studies Related to Outdoor Thermal-Acoustic Interactive Environments.

Authors	Year	Research Region	Acoustic Environment	Thermal Environment	Main Findings
Y. Jin, et al. [32]	2020	Simulated urban public space	Birdsong, dog bark, conversation, traffic, slow/fast music; SPL: 41.8–63.4 dB(A)	Winter (0°C), transitional seasons (20°C), summer (30°C)	Birdsong and slow music improved comfort in summer; loud or unpleasant sounds reduced comfort; low levels helped across seasons.
Y. Geng, et al. [33]	2022	Campus open spaces	Music, birdsong, water, wind/insects, crowd, machines (60–70 dB(A))	UTCI: Strong (>41.9°C), Moderate (22.2–41.9°C), None (12.3–22.2°C); seasonal: Summer 27.1°C, Winter 0.5°C	Music and natural sounds improved thermal comfort in cooler conditions; machine and crowd noise worsened perception in heat; lighting and visual comfort also influenced thermal responses.
Y. Chen, et al. [34]	2023	Urban park	Six sound types: water, birds, leaves, music, voices, construction; SPL: 52.9–77.8 dB(A)	UTCI: 31.35–38.92°C	High temperature and noise lowered comfort; natural sounds improved acoustic and thermal acceptance; sound-heat interaction intensified discomfort.
H. Chen, et al. [35]	2022	Campus open spaces in cold regions	Five sound types: music, water & birdsong, wind & insects, crowd, machinery; SPL: 60–70 dB(A)	Autumn: 10.4–20.1°C; Winter: −0.2–11.2°C	Pleasant sounds improved both thermal and acoustic comfort; neutral PET varied with sound type; thermal and acoustic comfort were positively related.
Y. Yin, et al. [3]	2022	Campus open spaces	SPL: 40–75 dB(A); sound sources: conversation, fountain, insects, wind, leaves	Autumn: 13.15–28.29°C; Winter: 10.10–47.28°C	Acoustic comfort matters most in extreme thermal conditions; moderate noise (60–70 dB(A)) yields best comfort; thermal factors dominate overall comfort.
M. Zhen, et al. [36]	2023	Outdoor campus spaces	SPL: 40–65 dB(A)	PET ranging from 8.2°C to 27.6°C (cool to slightly warm conditions)	Thermal comfort was influenced by PET; acoustic comfort by L_{Aeq}. Interaction effects varied by site. Best overall comfort at PET 18–23°C and L_{Aeq} ~ 50 dB(A).
K.K.-L. Lau, C.Y. Choi. [37]	2021	Urban open spaces	Dominated by traffic noise	Summer temperatures typically > 31°C, average ~27.8°C; evaluated using UTCI	Better acoustic and aesthetic quality improves outdoor thermal comfort and tolerance.
M. Du, et al. [38]	2023	Residential outdoor spaces	Five sound types: birdsong, music, opera, conversations, traffic; SPL: 60–80 dB(A)	PET: 13.80–46.16°C (neutral: 13.80–29.98°C)	High PET and traffic noise caused discomfort; thermal comfort dominated in summer, acoustic comfort in spring; comfort model had 84.7 % accuracy.
N. Mohammadzadeh, et al. [39]	2023	Urban parks	Measured SPL (L_{Aeq} approximately 40–60 dB(A)); soundscape included natural (vegetation), human, and traffic sounds	PET ~26.6°C average	Higher thermal discomfort reduces acoustic comfort; vegetation helps alleviate this combined discomfort.
M. Zhen, et al. [40]	2024	Urban plazas	SPL: 40.7–77.6 dB(A)	UTCI: 8.5–45.7°C	Natural sounds increased preferred temperature; traffic noise decreased it. Quiet environments made people more sensitive to heat, and vice versa.
W. Li, et al. [41]	2024	Urban pocket parks	SPL: 50.7–77.3 dB(A); sounds include birdsong, wind, traffic, voices	Air temperature: −5 °C to 27.8°C	Water-feature parks had highest comfort. High noise (>56 dB(A)) reduced thermal influence. Thermal, acoustic, and air quality effects interacted.
D. Lin, et al. [42]	2023	Outdoor public spaces	SPL: 30.0–81.7 dB(A); sources include traffic, voices, natural sounds	PET: 19.0–36.7°C; spring neutral PET ≈ 18.7°C	High noise reduced thermal comfort. PET affected sound sensitivity (~30°C optimal). Pleasant, low-stress settings improved overall comfort.
Y. Chen, et al. [43]	2024	Laboratory-simulated environments	bird calls, flowing water, human voices, light music, traffic noise, lawn mowing; each with 3 vol levels; SPL approximately 40–60 dB(A)	Simulated temperatures: 20°C (cool), 25°C (neutral), 30°C (hot)	Temperature and sound interact; moderate natural sounds enhance comfort; high temperature increases annoyance.

an average wind speed of 2.8 m/s [44].

This study was conducted in People's Park, Zhangjiakou (114°53′24″E, 40°49′33″N), where meteorological measurements and questionnaire surveys were carried out. Covering an area of 104,300 m^2, the park features diverse landscape elements, convenient accessibility, and is primarily frequented by local residents, making it an ideal site for investigating the interaction between thermal and acoustic comfort in urban parks of cold regions.

Urban sounds can be broadly categorized into three types based on their sources: natural sounds, human-generated sounds, and artificial sounds [45]. Considering the diversity of landscape features and thermal-acoustic environments, three representative spaces within the park were selected: the fitness activity area (P1), the ecological recreation area (P2), and the central waterscape area (P3). These areas exhibit distinct environmental characteristics. The ecological recreation area (P2) is predominantly shaded by trees, with natural sounds such as birdsong and insect chirping being the dominant auditory elements. The central waterscape area (P3) contains extensive water bodies with some vegetation and is located near urban roads, where artificial sounds such as traffic noise and construction machinery are prominent. In contrast, the fitness activity area (P1) has minimal tree cover, and the primary environmental sounds include human-generated noises such as conversation and exercise-related activities. Details of these selected spaces are presented in Fig. 1. These three areas reflect the typical spatial and acoustic heterogeneity of urban parks, providing a suitable basis for examining the interplay between thermal and acoustic environments and their impact on public perception and comfort.

2.2. Experimental design

2.2.1. Meteorological measurements

Field measurements were conducted in People's Park, Zhangjiakou, across three representative functional zones (P1, P2, P3) during three different periods: July (9th–10th), September (5th and 10th), and

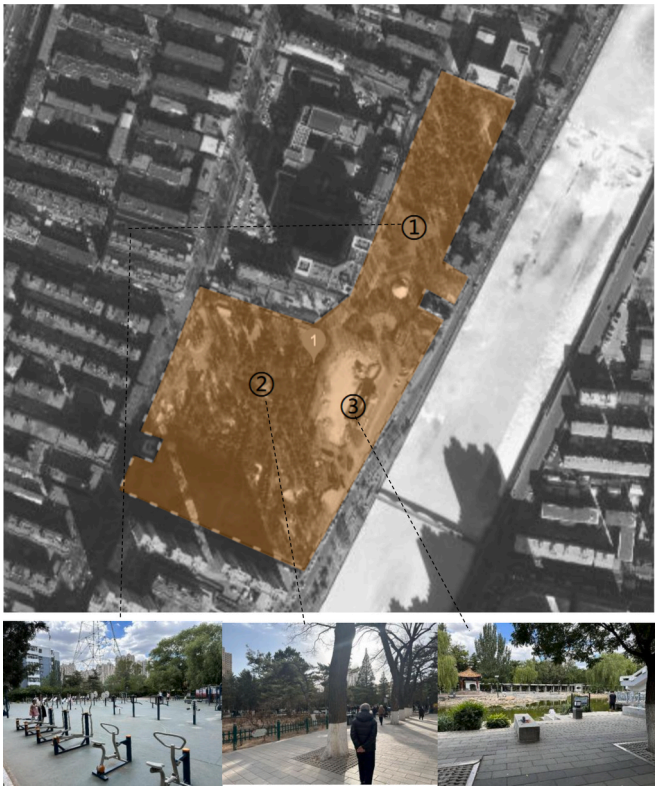


Fig. 1. Locations of the study and site configurations.

December (24th–25th) of 2023. Measurements were carried out daily from 09:00 to 17:00 to capture the main hours of public outdoor activity, and all surveys were conducted under typical sunny weather conditions to minimize the influence of extreme meteorological events.

To ensure temporal synchronization and data comparability across different sites, identical sets of meteorological and acoustic monitoring instruments were installed at each measurement point (see Table 2). The equipment was fixed in place throughout the measurement period and used to continuously record key environmental parameters, including air temperature (T_a), relative humidity (RH), wind speed (V_a), globe temperature (T_g), and sound pressure level (SPL).

Thermal environmental parameters were collected at one-minute intervals, while acoustic data were sampled every 10 s and summarized in 10-minute intervals as a full measurement cycle. To enhance the alignment between subjective evaluations and objective measurements, questionnaire surveys were conducted shortly before or after each hourly acoustic sampling cycle (e.g., 10:30, 11:30, etc.). Respondents were positioned within 2 m of the monitoring equipment to ensure consistency between their perceived thermal-acoustic environment and the measured data. Each questionnaire included precise time and location stamps, allowing for accurate matching of subjective responses with the corresponding microclimatic and acoustic conditions for subsequent model analysis.

Table 2
Instrument information.

Measurement	Instrument	Detectable Range	Resolution
Temperature	Yigu FT-SQ5 Handheld	−40–85 ℃	±0.3 ℃
Relative humidity	Weather Monitor	0–100 %RH	±3%RH
Air velocity		0–30 m/s	±0.3 m/s
Black-bulb temperature	Jianda Renke RS-HQ black sphere thermometer	−40–120 ℃	±0.1 ℃
Sound pressure level	BSWA309 multifunctional sound level meter	22 dBA–146 dBA 20 Hz–12.5 k Hz	±0.6–0.8 dBA

2.2.2. Questionnaire survey

This study conducted on-site surveys in three representative functional zones (P1, P2, and P3) of People's Park in Zhangjiakou. To ensure the consistency between subjective survey data and objective environmental measurements, all questionnaires were distributed and collected within the immediate vicinity of each monitoring site, with respondents positioned no more than 2 m from the instruments. Investigators approached passersby near the equipment and invited them to complete the questionnaires through face-to-face interaction. This approach ensured that participants' subjective evaluations reflected the same microclimatic and acoustic conditions being measured by the monitoring devices, thereby strengthening the reliability and explanatory power of the relationship between environmental variables and human perception.

As summarized in Table 3, the questionnaire consisted of the following components: (1) demographic and personal information, including gender, age, height, weight, clothing details (upper garment, lower garment, footwear), current activity status (e.g., sitting, standing, walking, exercising), and length of time spent at the survey location; (2) subjective environmental assessments, including thermal sensation vote (TSV, 9-point scale from −4 to +4), acoustic sensation vote (ASV), acoustic comfort vote (ACV), overall comfort vote (OCV), thermal comfort vote (TCV) (on a 7-point scale from −3 to +3 [46]). The surveys were conducted across three typical climatic conditions in 2023—summer, transitional season, and winter—resulting in a total of 798 valid responses (262 in summer, 335 in the transitional season, and 201 in winter). All data collection was carried out on a voluntary basis and did not involve any commercial use. The original questionnaire has been included in Appendix X to enhance the transparency and reproducibility of the research.

2.3. Environmental evaluation indicators

2.3.1. Thermal environment evaluation indicators

The Universal Thermal Climate Index (UTCI) was adopted in this study as the primary metric for evaluating outdoor thermal conditions [48]. Developed by the International Society of Biometeorology (ISB), the UTCI integrates multiple meteorological variables—including air temperature, relative humidity, wind speed, and mean radiant temperature—alongside individual physiological characteristics such as age, gender, height, weight, clothing insulation, and metabolic rate, to simulate the human body's comprehensive thermal stress response to outdoor environments.

To ensure temporal and spatial alignment between environmental conditions and subjective thermal perception, the date, exact time, and geographic coordinates of each questionnaire response were recorded during the survey. These inputs, along with the respondents' individual characteristics and real-time meteorological data (including air temperature, relative humidity, wind speed, and mean radiant temperature), were used as inputs for the Rayman model [49] to calculate the UTCI corresponding to each respondent's microclimatic exposure at the time of survey completion. This method enabled precise matching between objective environmental data and subjective thermal responses, thereby improving the accuracy and scientific validity of the model analysis.

Based on field measurements and model calculations, annual UTCI values ranged from −27.8 °C to 42.2 °C. Notably, no values within the 0 °C to 9 °C range were recorded, resulting in the representation of seven distinct thermal stress levels throughout the year, as summarized in Table 4. According to Table 5, no UTCI values were observed within the −40 °C to −27 °C range at sites P1 and P2, while no values between 9 °C and 26 °C were recorded at site P3. Specifically, the UTCI ranged from

Table 3
Subjective evaluations of the questionnaire survey.

	−4	−3	−2	−1	0	+1	+2	+3	+4
TSV	Very cold	Cold	Cool	Slightly cool	Neutral	Slightly warm	Warm	Hot	Very warm
	−3	−2	−1		0	+1		+2	+3
ASV	Very noisy	Noisy	Slightly noisy		Neutral	Slightly quiet		quiet	Very quiet
TCV	Very uncomfortable	uncomfortable	Slightly uncomfortable		Neutral	Slightly comfortable		Comfortable	Very comfortable
ACV	Very uncomfortable	uncomfortable	Slightly uncomfortable		Neutral	Slightly comfortable		Comfortable	Very comfortable
OCV	Very uncomfortable	uncomfortable	Slightly uncomfortable		Neutral	Slightly comfortable		Comfortable	Very comfortable

Table 4
UTCI ranges corresponding to different levels of heat stress [47].

Stress category	UCTI/℃	Stress category	UCTI/℃
Very strong heat stress	38–46	Slightly cold stress	0–9
Strong heat stress	32–38	Moderate cold stress	−13–0
Moderate heat stress	26–32	Strong cold stress	−27–13
No thermal stress	9–26	Very strong cold stress	−40–27

26.5 °C to 42.2 °C in summer, 24.5 °C to 41.5 °C in the transitional season, and −27.8 °C to −0.1 °C in winter.

2.3.2. Acoustic environment evaluation indicators

Given the discontinuous nature of sound sources at the measurement sites, this study utilized the equivalent continuous sound level (L_{Aeq}) as the primary acoustic environment evaluation metric. The daytime L_{Aeq} levels in the urban park ranged from 44.4 dB(A) to 71.2 dB(A) throughout the year. These values were classified into four categories in 10 dB increments: 40 dB(A)–50 dB(A), 50 dB(A)–60 dB(A), 60 dB(A)–70 dB(A), and above 70 dB(A).

2.3.3. Overall environmental evaluation indicators

Common indicators for overall environmental assessment include satisfaction level, annoyance level, and overall comfort. In this study, overall comfort was selected as the primary evaluation metric. A questionnaire survey was conducted among urban park users to assess their perceptions of overall comfort in the park environment.

2.4. Statistical analysis

Data preprocessing, including counting, classification, and cleaning, was conducted using Microsoft Excel. Subsequent statistical analyses were carried out using SPSS Statistics 27 (IBM), MATLAB, and Origin. A combination of correlation analysis, analysis of variance (ANOVA), and multiple regression analysis was applied to integrate and interpret both field measurements and questionnaire data.

Multiple linear regression models were developed to investigate the relationship between OCV and environmental variables across three seasonal contexts: summer, transitional seasons, and winter. The selected independent variables included the UTCI, equivalent continuous sound level (L_{Aeq}), TCV, and ACV. Model performance was assessed using the coefficient of determination (R^2), and multicollinearity diagnostics were conducted prior to regression modeling to ensure the validity of assumptions.

To assess the robustness and generalizability of the models, 10-fold cross-validation was performed. For each seasonal dataset, the data were randomly partitioned into 10 subsets. In each iteration, one subset was used for validation and the remaining nine for training. This process was repeated 10 times, and the mean and standard deviation of R^2 across the folds were computed to evaluate model stability.

3. Results

3.1. Descriptive analysis

3.1.1. Participant information

A total of 798 valid questionnaires were collected during the measurement periods, covering three representative climatic phases: summer (262 responses), the transitional season (335 responses), and winter (201 responses). Table 6 summarizes the demographic characteristics of respondents and their seasonal distribution. Among all participants, 465 were male (58.3 %) and 333 were female (41.7 %). The majority of respondents fell into two main age groups: 25–42 years (36.5 %) and over 60 years (32.8 %). Most participants (91.6 %) had lived in Zhangjiakou for more than one year, indicating a high degree of familiarity with the local outdoor environment.

In terms of spatial distribution, the number of questionnaires was relatively balanced across the three monitoring sites: 276 at site P1, 263 at site P2, and 259 at site P3. Each location was surveyed during all three seasonal periods to ensure a representative dataset that reflects varying thermal and acoustic environments across both time and space. Efforts were made to maintain balance in the number of questionnaires across different seasons and sites, while also achieving a sufficient total sample

Table 6
Basic information for the respondents.

Information	Classification	Frequency	Percentage
Gender	Male	465	58.3 %
	Female	333	41.7 %
Age	≤25 years	82	10.3 %
	25–42 years	291	36.5 %
	43–60 years	163	20.4 %
	≥60 years	262	32.8 %
Native of Zhangjiakou	Tourist	17	2.1 %
	Less than 1 year	50	6.3 %
	More than 1 year	731	91.6 %

Table 5
UTCI classification.

Stress category	Summer		Transitional season		Winter	
	Range/℃	Average/℃	Range/℃	Average/℃	Range/℃	Average/℃
Very strong heat stress	38.4–42.2	39.3	39.3–41.5	40		
Strong heat stress	32.1–37.7	34.2	33.1–37.8	35.9		
Moderate heat stress	26.5–32.0	30.6	26.5–31.9	29.4		
No thermal stress			24.5–25.5	25.1		
Moderate cold stress					−0.1–12.9	−8.4
Strong cold stress					−14.2–26.2	−20.3
Very strong cold stress					−27.8	−27.8

size. This strategy was intended to support reliable statistical analyses and spatial comparisons of subjective perception indices such as TSV, TCV, ASV, and ACV.

3.1.2. Thermal sensation vote (TSV) and thermal comfort vote (TCV)

As illustrated in Fig. 2(a), the distribution of TSV across the measurement points exhibits similar seasonal trends. During the transitional seasons, Point P2 recorded the highest proportion of neutral thermal sensation (TSV = 0) at 66.97 %, significantly exceeding that of P1 (48.21 %) and P3 (55.26 %), indicating that P2 provided a more thermally neutral environment during these periods. In summer, P2 also showed the lowest proportion of high-temperature sensations, with only 6.67 % of TSV = 2 and 15.56 % of TSV = 3, and no occurrences of TSV = 4. In contrast, the proportion of TSV = 3 at P1 and P3 reached 24.21 % and 22.08 %, respectively, suggesting that P2 offered relatively better thermal comfort under hot conditions. In winter, cold sensations were most pronounced at P1, where TSV = −3 and TSV = −4 accounted for 27.54 % and 4.35 %, respectively—higher than those observed at P3 and P2—indicating that P1 was more prone to inducing cold discomfort in colder seasons.

Fig. 2(b) presents the seasonal distribution of TCV. During the transitional seasons, Point P3 demonstrated the highest thermal comfort ratings, with 21.05 % of respondents selecting "comfortable" (TCV = 2) and 3.51 % selecting "very comfortable" (TCV = 3), significantly outperforming P1 (3.57 %) and P2 (8.26 %). This suggests that P3 provided a more satisfactory thermal environment during these periods. In summer, P2 had the lowest proportion of discomfort votes, with only 2.22 % for TCV = −2 and no occurrences of TCV = −3. In contrast, both P1 and P3 showed higher proportions in these categories, indicating that P2 was more conducive to maintaining thermal comfort under hot conditions. In winter, a different trend emerged: P3 recorded the highest proportion of "very uncomfortable" votes (TCV = −3) at 14.71 %, compared to 5.80 % for P1 and 3.13 % for P2, reflecting reduced thermal comfort at P3 under cold conditions.

3.1.3. Acoustic sensation vote (ASV) and acoustic comfort vote (ACV)

As shown in Fig. 3(a), during the transitional seasons, Point P1 had the highest proportion of "noisy" perceptions (ASV = –2) at 27.68 %, whereas P2 recorded the greatest proportion of "neutral" acoustic sensations (ASV = 0) at 52.39 %. This indicates that the acoustic environment at P1 was relatively poor during these periods, while P2 offered a quieter setting. In summer, the proportion of neutral acoustic sensation at P2 increased to 62.22 %, significantly higher than that at P1 (30.53 %) and P3 (46.75 %). P1 and P3 exhibited similar levels of negative acoustic perception, with comparable proportions for "noisy" (ASV = –2) and "very noisy" (ASV = –3), while P2 showed no reports of either. These findings suggest that P2 provided a more pleasant acoustic environment in hot weather. In winter, however, the trend reversed: P3 had the highest proportions of "quiet" (ASV = 2) and "very quiet" (ASV = 3), at 7.35 % and 2.94 %, respectively, indicating that P3 offered the quietest acoustic environment during colder conditions.

Fig. 3(b) further illustrates the distribution of ACV. During the transitional seasons, Point P1 exhibited the highest proportion of participants reporting acoustic discomfort (ACV = –2) at 16.07 %, indicating a pronounced level of dissatisfaction with the acoustic environment. In comparison, the corresponding values for P2 and P3 were notably lower, at 9.17 % and 9.65 %, respectively. In summer, P2 recorded the highest percentage of neutral acoustic comfort (ACV = 0) at 63.33 %, surpassing P3 (40.26 %) and P1 (32.63 %), reinforcing the observation that P2 provided a more favorable acoustic environment

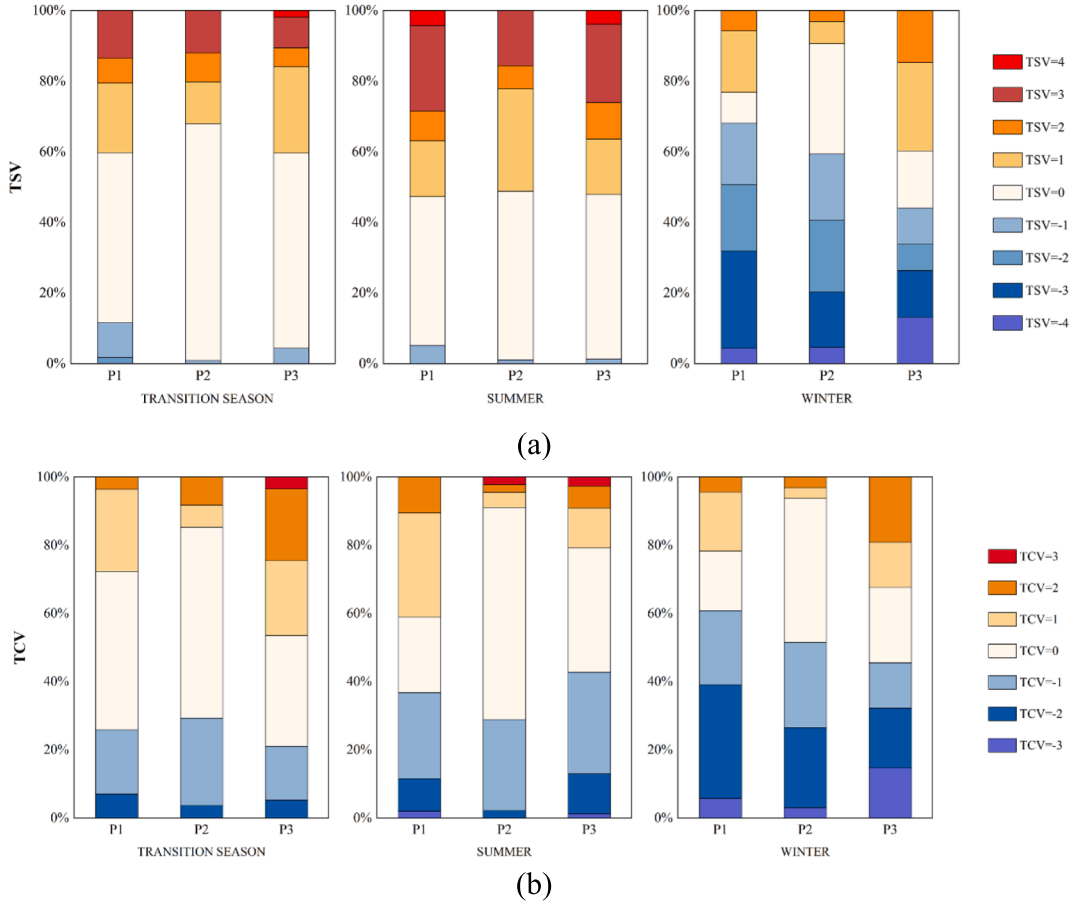


Fig. 2. Distribution of the TSV (a) and TCV (b) among different sites.

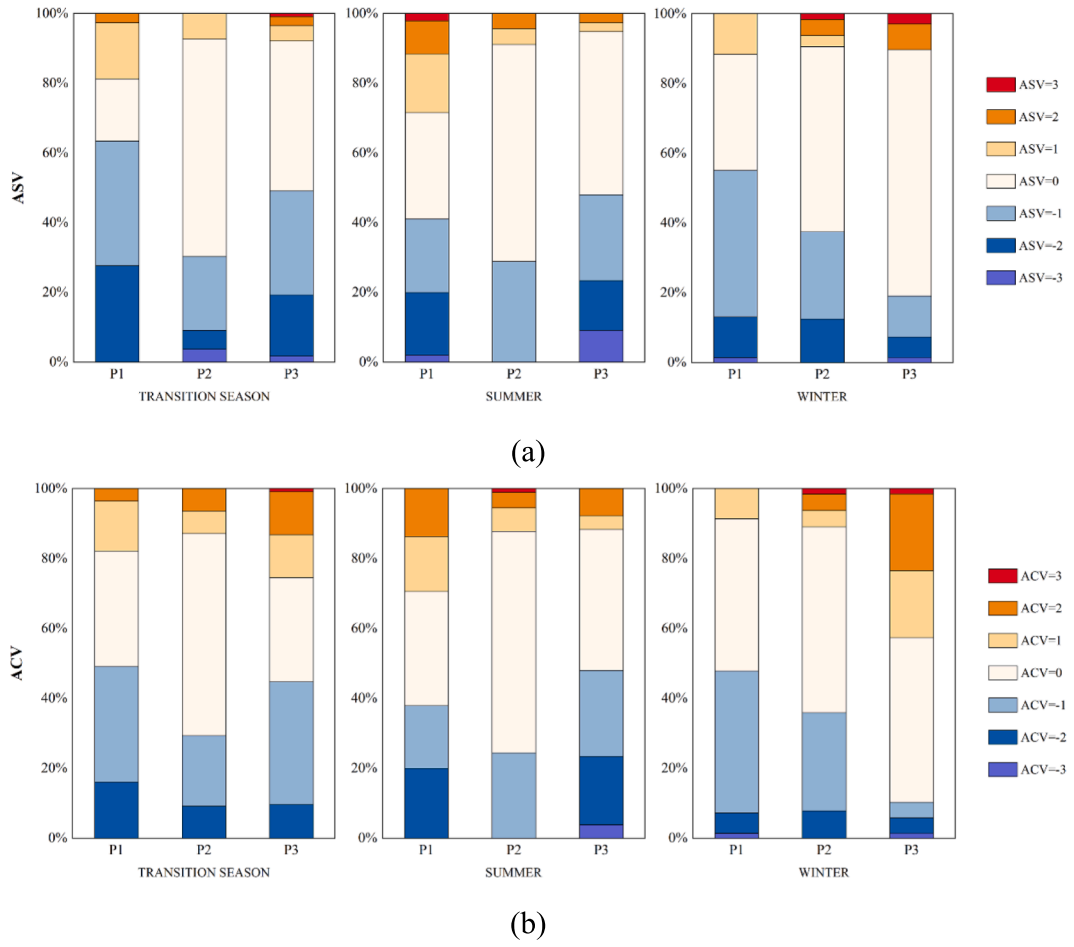


Fig. 3. Distribution of the ASV (a) and ACV (b) among different sites.

during hot weather. In winter, P3 achieved the highest ratings for acoustic comfort, with 22.06 % of respondents selecting "comfortable" (ACV = 2) and 1.47 % selecting "very comfortable" (ACV = 3), substantially exceeding the values observed at P1 and P2. These results suggest that P3 offered the best acoustic comfort under cold seasonal conditions.

3.1.4. *Overall comfort votes (OCV)*

As shown in Fig. 4, during the transitional seasons, the three measurement points exhibited relatively similar proportions of discomfort perception (OCV = –1 and –2), ranging between 22 % and 28 %. Point P2 recorded the highest proportion of "neutral" responses (62.39 %), while P3 had the greatest combined share of "slightly comfortable" and "comfortable" votes (OCV = 1 and 2), totaling 30.70 %. These findings suggest that P3 offered the most favorable overall environmental conditions during transitional periods.

In summer, P2 again stood out with the highest proportion of "neutral" evaluations (74.44 %), significantly exceeding those of P1 (30.53 %) and P3 (48.05 %), indicating a more balanced integration of thermal and acoustic environments at this location. However, P1 had the highest proportion of positive comfort responses (36.84 % for OCV = 1 and 2), suggesting that its environment was perceived as more ideal during hot conditions. In contrast, P3 received the highest level of discomfort evaluations (OCV = –1 and –2), accounting for 38.96 %.

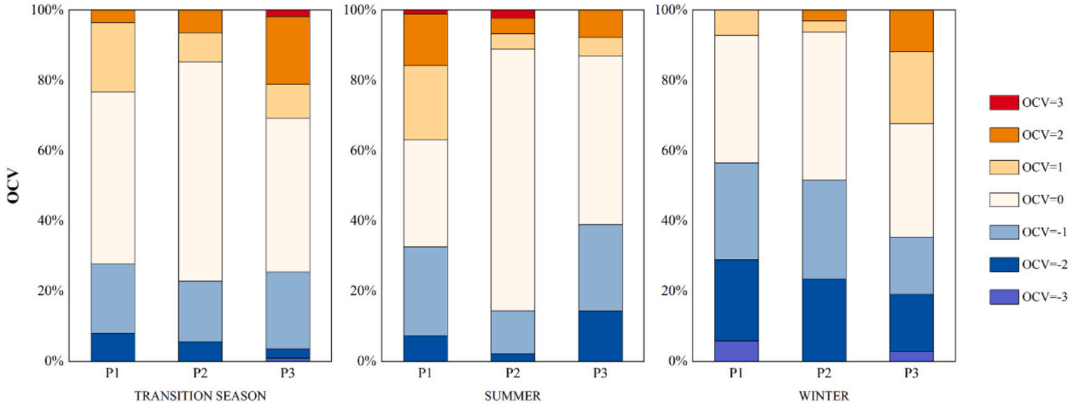


Fig. 4. Distribution of the OCV among different sites.

In winter, P1 registered the highest combined proportion of "uncomfortable" responses (56.53 %), indicating that it provided the least favorable environmental conditions during cold weather. Conversely, P3 had the highest share of "slightly comfortable" and "comfortable" evaluations, totaling 32.35 %, suggesting superior overall comfort at this location in winter. The proportion of "neutral" responses was relatively similar across all three points, ranging from 32 % to 42 %, reflecting a more dispersed pattern of environmental perception in the colder season.

3.2. Interaction between UTCI and sound types

3.2.1. Effects of thermo-acoustic interaction on thermal perception

As shown in Fig. 5(a), TSV generally increased with rising UTCI across all sound types. At each UTCI level, sound type had a significant effect on TSV. Under both cold and heat stress conditions, artificial sounds significantly elevated individuals' thermal sensation. Specifically, in cold stress conditions, artificial sounds resulted in TSV values that were 0.71–0.975 higher than those associated with other sound types; under heat stress, this difference ranged from 0.21 to 0.52. These findings indicate that artificial sounds can intensify thermal perception in hot environments while mitigating cold perception in low-temperature settings, with the cold perception mitigation effect being more pronounced than the enhancement of heat perception. Under moderate heat stress, the TSV associated with artificial sounds was 0.36 higher than that of natural sounds, and under strong heat stress, the difference increased to 0.57. This suggests that natural sounds may help alleviate thermal discomfort in hot environments, with their moderating effect becoming more evident as temperature rises.

As illustrated in Fig. 5(b), under moderate and strong heat stress, the TCV associated with artificial sounds was approximately 0.4 lower than that of natural sounds, indicating that artificial sounds significantly reduce subjective thermal comfort in hot conditions. In contrast, natural sounds significantly improved TCV under moderate, strong, and very strong heat stress, with corresponding TCV values of 0.39, −0.09, and 0.43, respectively. These results highlight the potential of natural sounds to enhance thermal comfort in hot environments. In cold conditions, human-generated sounds had a negative impact on TCV, with values of −1.64 and −0.64 under moderate and strong cold stress, respectively, indicating that such sounds may reduce thermal comfort in cold environments.

As shown in Table 7(a), the correlations between thermal variables

Table 7a
Correlation Analysis between TSV and ACV, and between TCV and ACV across Seasons.

Season	Summer		Transitional season		Winter	
	TSV	TCV	TSV	TCV	TSV	TCV
ACV	0.094	−0.093	**0.257****	**−0.215****	0.109	0.050

Note: **p < 0.01, *p < 0.05 represents statistically significant difference.

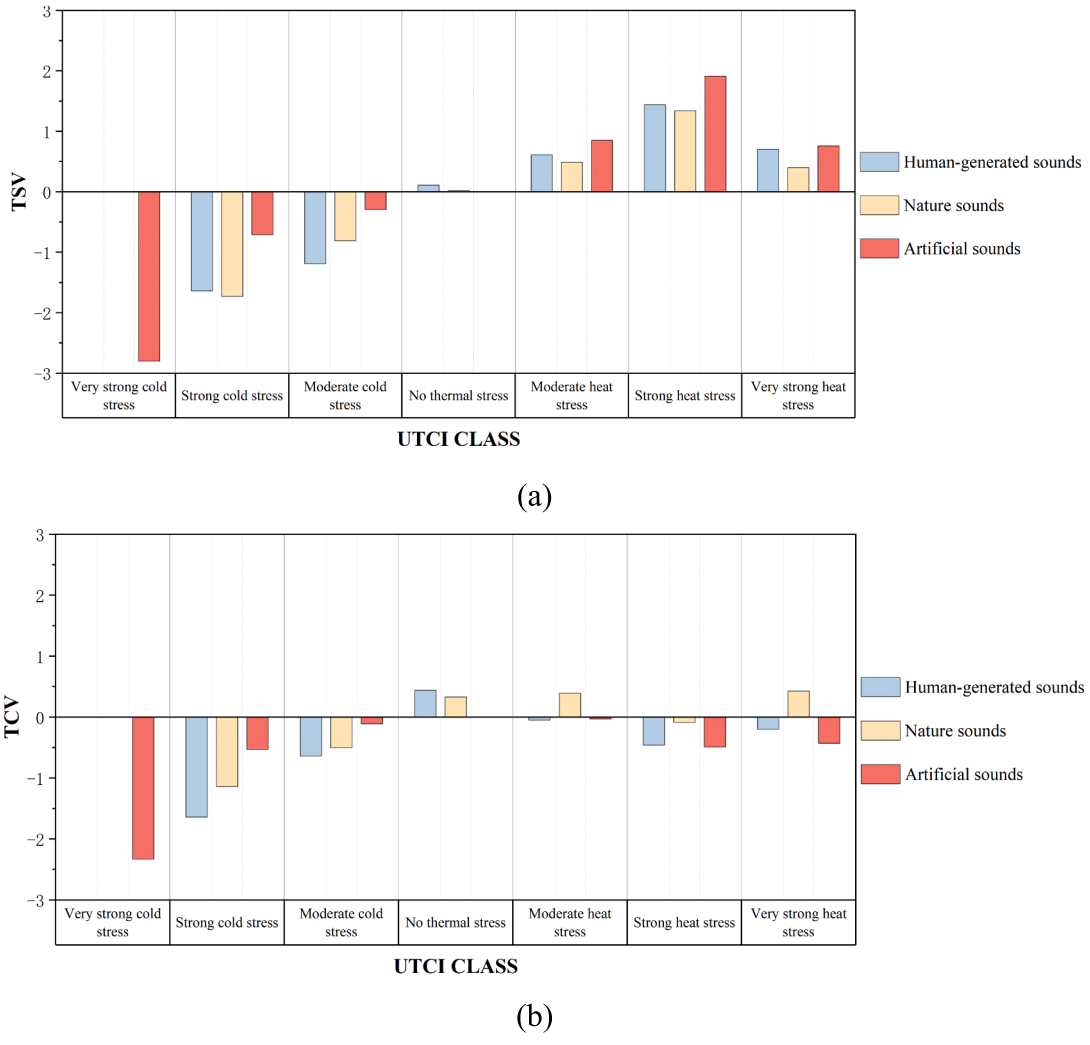


Fig. 5. Relationship between mean TSV and UTCI (a), TCV and UTCI (b) among sound types.

(TSV, TCV) and ACV were weak and statistically insignificant in both summer and winter, with coefficients close to zero. Although these near-zero results indicate the absence of a measurable relationship under extreme hot or cold conditions, they are still valuable in defining the boundary conditions of thermal–acoustic interactions—suggesting that when thermal stress dominates perception, the acoustic environment plays only a secondary role. In contrast, in the transitional season, TCV was significantly and negatively correlated with ACV, while TSV showed a significant positive correlation, indicating that under milder thermal conditions, a favorable acoustic environment is more closely linked to improved thermal perception. The following section examines how TSV and TCV respond to changes in ACV under different UTCI-based thermal stress levels, providing a cross-season perspective beyond the seasonal grouping shown in Table 7(a).

As shown in Fig. 6(a), when ACV remains constant, TSV increases significantly with rising UTCI, indicating a strong effect of environmental heat stress on thermal sensation. For instance, when ACV = 0, TSV increases progressively from −2.9 under very strong cold stress to 1.02 under strong heat stress, showing a clear linear upward trend. Under the same UTCI level, the influence of ACV on TSV varies depending on the degree of thermal stress. Under moderate heat stress, TSV reaches its minimum value (0.05) at ACV = 3, which is 0.69 lower than the maximum TSV (0.72) observed at the same thermal level. This suggests that a comfortable acoustic environment can help mitigate thermal discomfort. However, under strong heat stress, the response of TSV to ACV becomes less pronounced, with a narrower variation range (1.02 to 1.44), indicating that in extreme heat, the regulatory effect of sound on thermal perception diminishes.

As shown in Fig. 6(b), TCV reaches its highest value under no thermal stress and decreases as the environment becomes either colder or hotter. Across all levels of thermal stress, significant differences in TCV are observed under different ACV levels, indicating that the acoustic environment plays a crucial role in thermal comfort perception. When ACV = 2, the corresponding TCV values under strong cold stress, moderate cold stress, no thermal stress, moderate heat stress, and strong heat stress are 0.18, 1.57, 2.00, 1.89, and 0.75, respectively. These values are higher than the lowest TCV observed under each respective thermal condition by 2.18, 3.21, 1.95, 2.08, and 1.42, respectively. This demonstrates that a favorable perception of the acoustic environment can enhance thermal comfort, regardless of whether the thermal environment is cold or hot. Moreover, the range of TCV variation under cold conditions (e.g., moderate and strong cold stress) is considerably larger than that under hot conditions (e.g., moderate and strong heat stress), suggesting that individuals are more sensitive to acoustic environments in colder settings.

3.2.2. *Effects of thermo-acoustic interaction on acoustic perception*

As shown in Fig. 7(a), sound type has a significant effect on ASV.

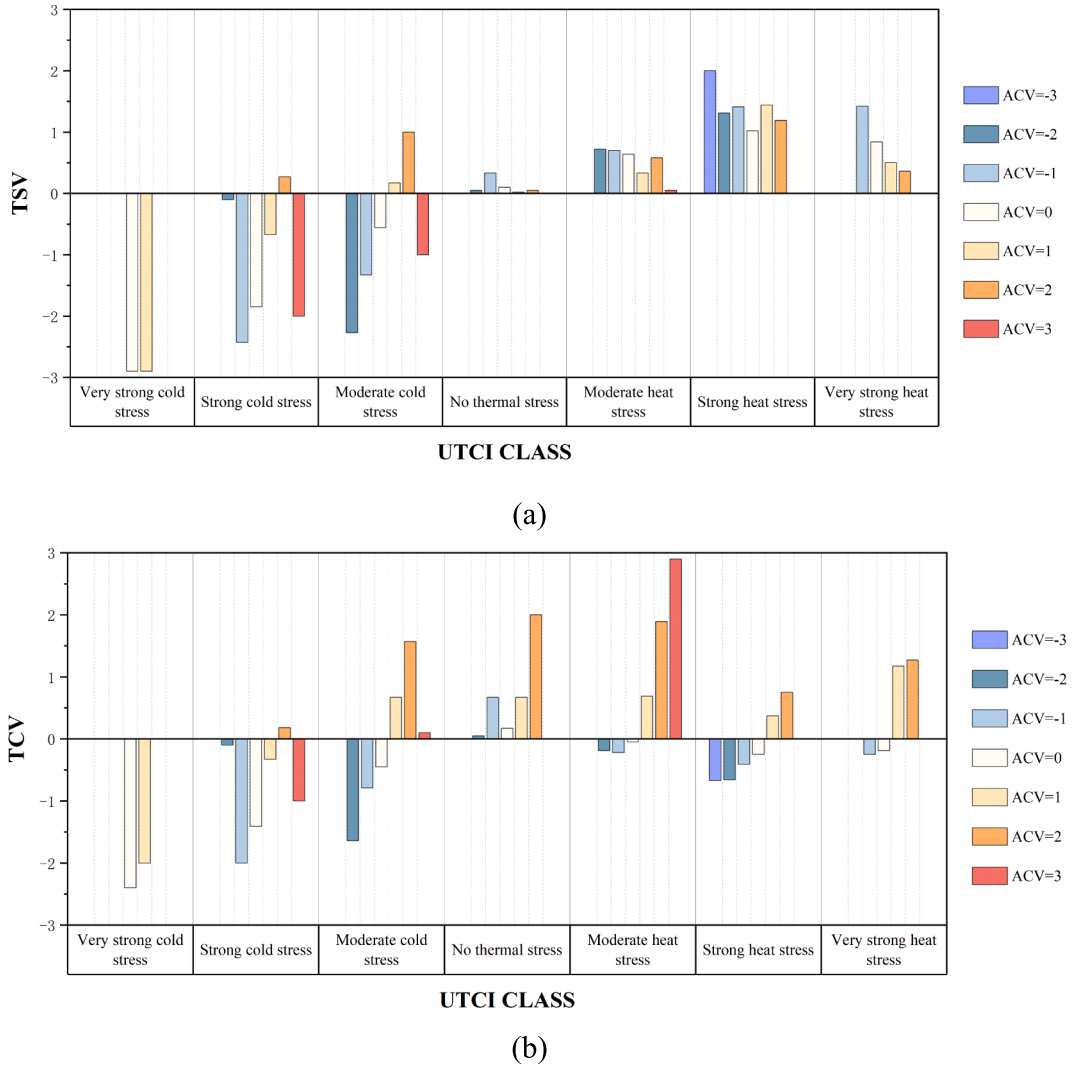


Fig. 6. Relationship between mean TSV and UTCI (a), TCV and UTCI (b) among ACV levels.

Overall, across all UTCI levels, natural sounds produced the highest mean ASV (0.01), significantly outperforming both human-generated sounds (−0.46) and artificial sounds (−0.48). Under moderate and strong cold stress, the ASV for natural sounds reached 0.42 and 0.23, respectively—substantially higher than those for the other sound types. This suggests that in cold environments, individuals exhibit greater acceptance of natural sounds, while showing stronger aversion to artificial and human-generated sounds. As thermal stress increased, the ASV differences among the three sound types gradually diminished, indicating a reduced influence of sound type on subjective noise acceptance in hotter conditions. From a broader perspective, ASV for all sound types exhibited a decreasing trend from very strong cold stress to very strong heat stress. Simultaneously, the inter-type differences in ASV narrowed, indicating that as ambient temperature rises, individuals' tolerance for sound generally declines, and ASV becomes more susceptible to thermal environmental influences.

As shown in Fig. 7(b), ACV was primarily influenced by sound type. The average ACV values from highest to lowest were: natural sounds (0.40), artificial sounds (−0.29), and human-generated sounds (−0.30). Natural sounds resulted in the highest ACV under no thermal stress and moderate heat stress, both exceeding 1.08. However, under extreme thermal stress—such as strong cold stress and strong heat stress—ACV values for natural sounds dropped notably to 0.27 and −0.27, respectively. This indicates that natural sounds are more likely to be perceived as comfortable in thermally favorable environments, but their positive effects diminish when the thermal environment deviates significantly from neutrality. Although sound type was the dominant factor affecting ACV, thermal stress also introduced noticeable fluctuations, suggesting that the thermal environment plays a regulatory role in subjective perception of the acoustic environment.

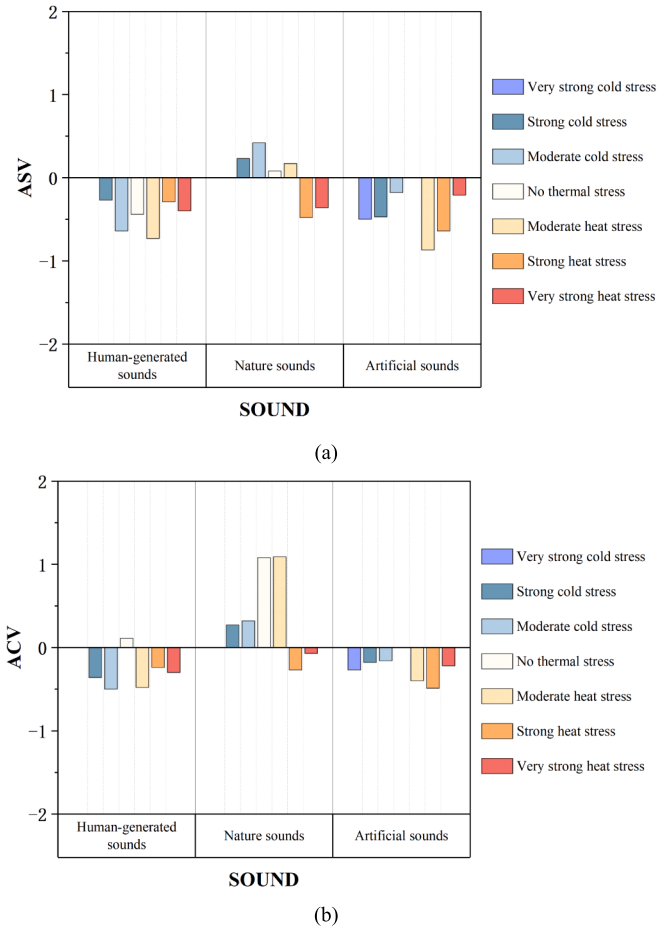


Fig. 7. Relationship between mean ASV and sound types (a), ACV and sound types (b) among UTCI classes.

As shown in Table 7(b), the correlations between TCV and either ASV or ACV were weak and statistically insignificant in both summer and winter, with coefficients close to zero. Although these near-zero results indicate the absence of measurable associations under extreme thermal conditions, they remain informative for defining the limits of thermal–acoustic interactions, suggesting that when heat or cold stress strongly dominates human perception, the influence of acoustic factors on thermal comfort is minimal. In the transitional season, however, TCV showed a significant negative correlation with both ASV and ACV, implying that in milder thermal environments, better acoustic evaluations are linked to higher thermal comfort. The following section explores how ASV and ACV vary with TCV under different sound types, offering further insight into the interaction between thermal and acoustic perceptions.

As shown in Fig. 8(a), in environments with human-generated sounds and artificial sounds, ASV generally increased with rising TCV, indicating a trend in which higher thermal comfort is associated with greater subjective acceptance of the acoustic environment. For example, under artificial sound conditions, ASV decreased from 0.5 to −0.73 as TCV dropped from 3 to −3, suggesting that participants evaluated the surrounding soundscape more positively when they felt thermally comfortable. In contrast, under natural sound conditions, a clear threshold pattern was observed in the relationship between ASV and TCV: when TCV $>$ 0, ASV was positive; when TCV $\leq$ 0, ASV became negative. This indicates a strong alignment between thermal and acoustic comfort in the presence of natural sounds—when thermal comfort improves, individuals are more likely to perceive the acoustic environment as quiet or pleasant, whereas thermal discomfort tends to coincide with perceptions of noisiness.

Fig. 8(b) further supports this association. In the presence of natural sounds, ACV increased with TCV, particularly under higher thermal comfort levels. When TCV equaled 1, 2, and 3, ACV reached 0.85, 1.31, and 1.90, respectively—substantially higher than those observed under other thermal sensation conditions. In contrast, when TCV $\leq$ 0, ACV values for all sound types remained below 0 and exhibited limited variability. This pattern indicates that under thermally uncomfortable conditions, perceived acoustic comfort tends to be uniformly low, highlighting the positive regulatory role of thermal comfort in shaping subjective evaluations of the acoustic environment.

3.3. Interaction between UTCI and sound pressure level

3.3.1. Influence of thermal-acoustic interaction on thermal perception

Analysis of variance (ANOVA) revealed significant main and interaction effects. Specifically, as shown in Table 8(a), both UTCI and L_{Aeq} significantly influenced TSV across all three seasons ($p < 0.01$).

As illustrated in Fig. 9(a), TSV demonstrates a clear seasonal trend in response to changes in UTCI. In summer, TSV values predominantly range from 0 to 4.0. When UTCI exceeds 39.3 °C, thermal sensation shifts toward hot; within the 30.6–39.3 °C range, sensations are warm, and approach neutrality near 30.6 °C. During the transitional seasons, TSV values span from −1.5 to 3.5, indicating a shift from neutral to slightly warm or hot sensations. In contrast, TSV in winter ranges mainly from −4.0 to 1.0, with cold sensations occurring when UTCI drops below −20.3 °C and cool sensations between −8.4 °C and −20.3 °C. Under the

Table 7b
Correlation Analysis between ASV and TCV, and between ACV and TCV across Seasons.

Season	Summer		Transitional season		Winter	
	ASV	ACV	ASV	ACV	ASV	ACV
TCV	−0.070	−0.093	**−0.133***	**−0.215****	0.063	0.050

Note: **p < 0.01, *p < 0.05 represents statistically significant difference.

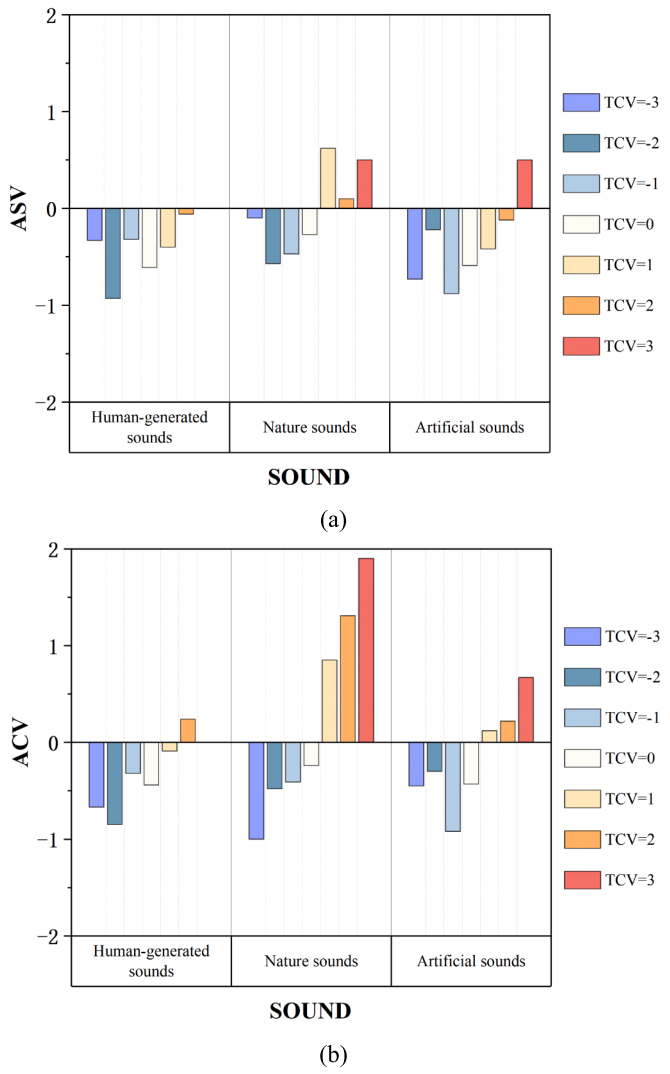


Fig. 8. Relationship between mean ASV and sound types (a), ACV and sound types (b) among TCV levels.

same acoustic environment (L_{Aeq}), TSV consistently increases with UTCI across all seasons, highlighting the dominant role of thermal conditions in determining thermal sensation. The average TSV range (the difference between the maximum and minimum) across seasons is 2.6 in summer, 4.8 in the transitional seasons, and 2.6 in winter, indicating that thermal sensitivity is highest during transitional periods. In terms of the acoustic environment, seasonal variations are observed in response to L_{Aeq}. In summer, at constant UTCI levels, TSV increases slightly with rising L_{Aeq}; for example, when noise levels rise from 40–60 dB(A) to 60–70 dB(A), TSV increases by approximately 0.5 on average. In contrast, under the same sound levels in winter, TSV decreases by about 0.6, suggesting that noise may amplify cold sensations. These findings indicate that sound does not only affect acoustic comfort but can also intensify thermal discomfort caused by extreme heat or cold, with amplification effects more pronounced in summer and winter.

According to Table 8(a), ANOVA revealed that both UTCI and L_{Aeq} had a statistically significant effect on TCV in all three seasons ($p < 0.01$).

As shown in Fig. 9(b), the impact of different thermo-acoustic environments on TCV varies significantly across seasons. In summer, TCV mainly ranges from −3.0 to 0.5, with thermal comfort decreasing as UTCI increases. Specifically, when UTCI exceeds 39.3 °C, participants report feeling very uncomfortable; at 34.2 °C, they feel slightly uncomfortable; and at 30.6 °C, they experience slightly comfortable conditions. During the transitional seasons, TCV spans a wider range (from −2.0 to 2.0), indicating varying degrees of comfort. Participants report feeling comfortable within the 25.1–29.4 °C range, slightly uncomfortable from 29.4–35.9 °C, and very uncomfortable when UTCI exceeds 35.9 °C. In winter, TCV is concentrated between −3.0 and −1.0; the strongest discomfort occurs when UTCI falls below −20.3 °C, and slightly uncomfortable conditions are observed around −8.4 °C. The effect of thermal conditions on TCV also exhibits distinct seasonal trends. In both summer and the transitional seasons, TCV declines with increasing UTCI, with average ranges of 2.9 and 4.0, respectively, indicating that high temperatures significantly reduce thermal comfort. In contrast, during winter, TCV slightly increases with rising UTCI, with an average range of 1.7, suggesting that a reduction in cold stress improves comfort perception. Acoustic conditions further modulate TCV under the same thermal environment. In summer, TCV is lowest when L_{Aeq} is within 60–70 dB(A), with an average range of 0.4. During the transitional seasons, the lowest TCV occurs at 40–50 dB(A), with an average range of 0.8. In winter, TCV increases as L_{Aeq} decreases, with an average range of 0.4. These findings suggest that both excessively high and low sound levels may undermine thermal comfort perception. Therefore, optimizing the acoustic environment plays a crucial role in enhancing overall thermal comfort.

3.3.2. *Influence of Thermal-Acoustic interaction on acoustic perception*

The ANOVA results (Table 8b) demonstrated that acoustic conditions, as measured by L_{Aeq}, significantly affected ASV in all three seasons ($p < 0.01$). In contrast, thermal conditions represented by UTCI had a significant impact on ASV only in summer ($p < 0.01$), with no statistically significant effects observed during the transitional season or winter ($p > 0.05$).

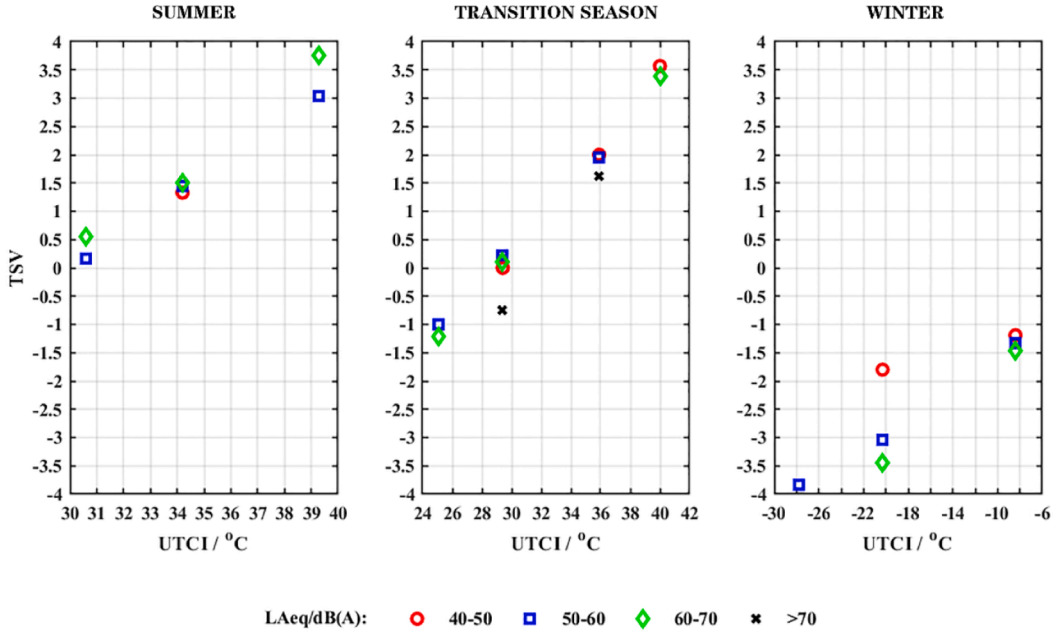
As shown in Fig. 10(a), the impact of the acoustic environment on ASV is pronounced across all seasons. Overall, ASV significantly decreases with increasing L_{Aeq}, indicating that higher noise levels lead to a greater perception of noisiness. The average range of ASV variation is 2.3 in summer, 3.0 in the transitional seasons, and 2.9 in winter, underscoring the strong influence of sound levels on subjective acoustic comfort regardless of season. The effect of thermal conditions on ASV is particularly evident in summer. At the same L_{Aeq} levels, ASV in summer fluctuates markedly with changes in UTCI. The lowest ASV occurs at a UTCI of 30.6 °C, with an average range of 0.7, suggesting heightened auditory sensitivity under mild heat stress. In contrast, during transitional seasons and winter, the ASV response to UTCI variation is minimal, with average ranges of only 0.1 and 0.2, respectively. This indicates that individuals' subjective evaluations of the acoustic environment remain relatively stable under these conditions, with ASV values primarily distributed between −2.0 and 2.5.

According to Table 8(b), ANOVA results for ACV revealed that L_{Aeq}

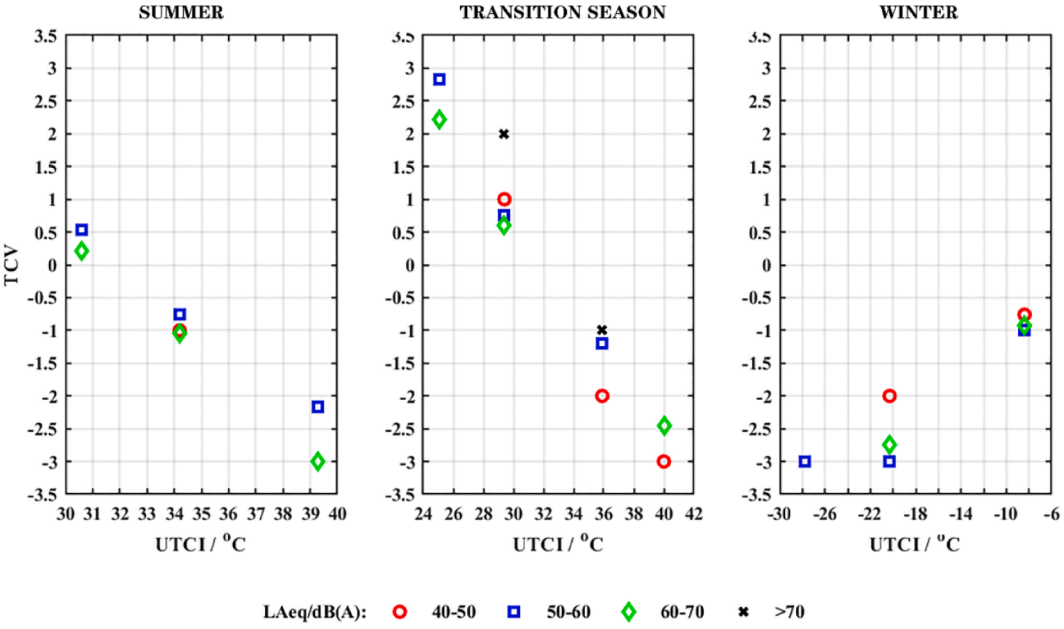
Table 8a
Results of ANOVA for the Effects of UTCI and L_{Aeq} on TSV and TCV in Different Seasons.

	Summer		Transitional season		Winter	
	UTCI	LAeq	UTCI	LAeq	UTCI	LAeq
TSV	**<0.001*****	**<0.001*****	**<0.001*****	**<0.001*****	**<0.001*****	**<0.001*****
TCV	**<0.001*****	**<0.001*****	**<0.001*****	**<0.001*****	**<0.001*****	**<0.001*****

Note: *** indicates statistical dependence significant at the 0.001 level; ** at the 0.01 level; and * at the 0.05 level.



(a)



(b)

Fig. 9. The values of TSV(a), TCV(b) under different conditions of the thermal-acoustic environment.

Table 8b
Results of ANOVA for the Effects of UTCI and L_{Aeq} on ASV, and ACV in Different Seasons.

	Summer		Transitional season		Winter	
	UTCI	LAeq	UTCI	LAeq	UTCI	LAeq
ASV	**0.008****	**<0.001*****	0.886	**<0.001*****	0.066	**<0.001*****
ACV	**<0.001*****	**<0.001*****	0.993	**<0.001*****	**<0.001*****	**<0.001*****

Note: *** indicates statistical dependence significant at the 0.001 level; ** at the 0.01 level; and * at the 0.05 level.

had a significant main effect across all three seasons ($p < 0.01$), while UTCI significantly influenced ACV in both summer and winter ($p < 0.01$).

As depicted in Fig. 10(b), the influence of the acoustic environment on ACV is also significant in all seasons. In general, ACV decreases as L_{Aeq} increases, demonstrating that higher noise levels reduce perceived

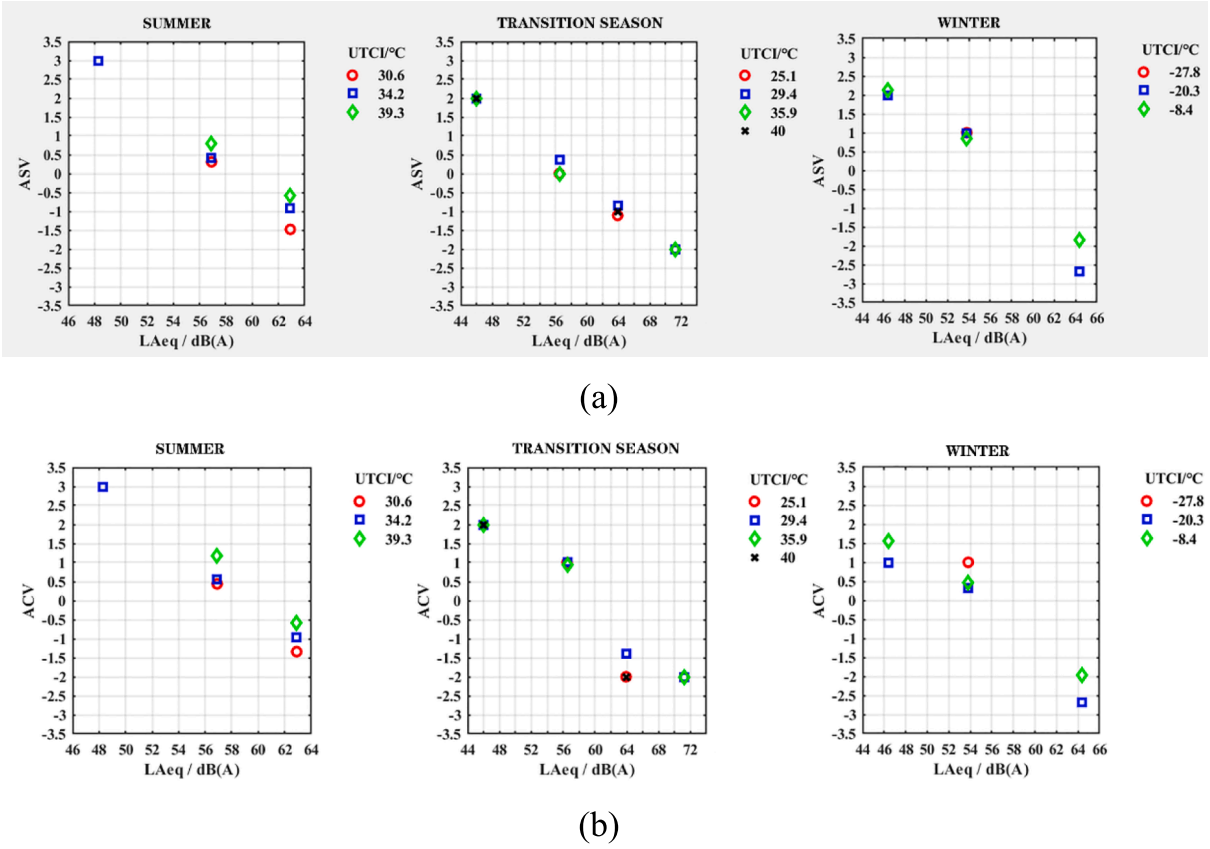


Fig. 10. The values of ASV (a), ACV (b) under different conditions of the thermal-acoustic environment.

acoustic comfort. The average range of ACV is 2.5 in summer, 3.7 in the transitional seasons, and 3.6 in winter, reflecting notable seasonal variability. However, it is worth noting that when L_{Aeq} exceeds 60 dB (A), ACV tends to stabilize, and further increases in noise level do not result in a significant decline in acoustic comfort. The regulatory effect of thermal conditions on ACV is most evident in summer and winter. In summer, under constant sound levels, ACV increases with rising UTCI—from 30.6 °C to 39.3 °C, the difference reaches 0.7—indicating that mild heat stress amplifies sensitivity to sound and enhances discomfort. A similar trend is observed in winter, where ACV increases by 0.5 as UTCI rises from −27.8 °C to −8.4 °C, suggesting that intense cold stress may also exacerbate auditory discomfort. In contrast, during transitional seasons, ACV shows minimal variation under different UTCI conditions, implying a relatively stable perception of the acoustic environment under thermally neutral conditions.

3.4. *Influence of thermal-acoustic interaction on overall comfort*

As shown in Table 8(c), ANOVA revealed statistically significant main and interaction effects of UTCI and L_{Aeq} on OCV across all three seasons ($p < 0.01$).

As shown in Fig. 11, OCV is influenced by both thermal and acoustic environments. Regarding thermal conditions, in summer and transitional seasons, OCV decreases as UTCI increases, indicating that elevated temperatures reduce overall comfort. In contrast, during winter, OCV increases with rising UTCI (i.e., decreasing cold stress), suggesting improved thermal comfort under milder cold conditions. With respect to the acoustic environment, OCV consistently declines with increasing L_{Aeq} at the same UTCI level. The average range of OCV variation across seasons is 1.7 in summer, 1.6 in the transitional seasons, and 1.8 in winter, demonstrating that elevated sound levels have a significant adverse effect on overall comfort. However, once L_{Aeq} exceeds 60 dB(A), OCV tends to stabilize, indicating a diminishing marginal effect of sound intensity on perceived comfort.

In terms of thermal-acoustic interaction, seasonal differences are evident. In summer and transitional seasons, the difference in OCV across UTCI levels narrows as L_{Aeq} increases. For instance, in summer, the average OCV range decreases from 0.8 at 50–60 dB(A) to 0.4 at 60–70 dB(A); similarly, in transitional seasons, the average range reduces progressively from 1.0 at 40–50 dB(A) to 0.5 above 70 dB(A). This trend suggests that high noise levels may mask the differential impact of thermal stress on overall comfort during warmer periods. Conversely, in winter, the opposite pattern is observed. As L_{Aeq} increases, the influence of UTCI on OCV becomes more pronounced, with the average range expanding from 0.2 to 1.2. This indicates that under cold conditions, elevated noise levels may exacerbate thermal discomfort, highlighting a stronger amplification effect of noise on thermal perception during winter.

Table 8c
Results of ANOVA for the Effects of UTCI, L_{Aeq}, and Their Interaction on OCV in Different Seasons.

	Summer			Transitional season			Winter		
	UTCI	L_{Aeq}	UTCI*L_{Aeq}	UTCI	L_{Aeq}	UTCI*L_{Aeq}	UTCI	L_{Aeq}	UTCI*L_{Aeq}
OCV	**<0.001*****	**<0.001*****	**<0.001*****	**<0.001*****	**<0.001*****	**<0.001*****	**<0.001*****	**<0.001*****	**<0.001*****

Note: *** Dependence is significant at 0.001 level. ** Dependence is significant at 0.01 level. * Dependence is significant at 0.05 level.

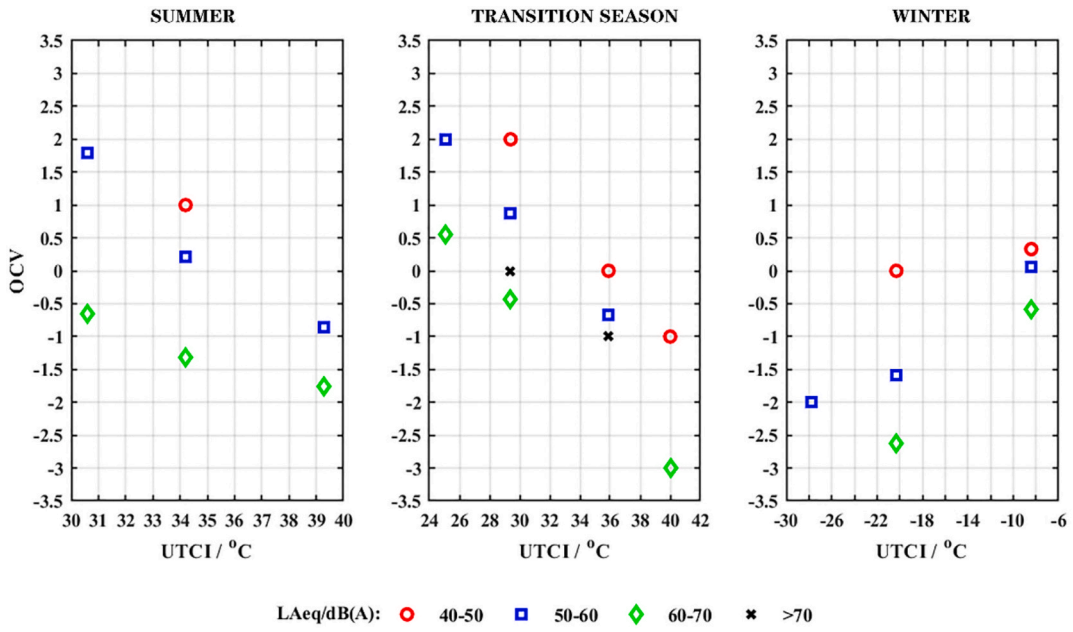


Fig. 11. The values of OCV under different conditions of the thermal-acoustic environment.

4. Discussion

4.1. Combined effects of temperature and sound type

Our study found that sound type significantly influenced TSV and TCV. Artificial sounds intensified subjective TSV in hot environments while mitigating the perception of cold in cold environments, consistent with previous findings [33]. In hot conditions, natural sounds alleviated subjective TSV and enhanced thermal comfort. Jin et al. [32] observed that bird songs significantly reduced TSV in hot environments, which aligns with our results. Sound type also had a significant impact on TCV. In hot environments, artificial sounds decreased subjective TCV, exacerbating thermal discomfort, whereas natural sounds increased subjective TCV, alleviating thermal discomfort. These findings are consistent with those of Jin et al. [32,50], possibly because artificial sounds are often associated with noisy, heat-intensive work environments, amplifying discomfort, whereas natural sounds evoke images of cooling breezes, thereby reducing thermal distress.

ACV significantly affected subjective TSV and TCV. In both warm and hot environments, a comfortable soundscape reduced TSV. Regardless of thermal conditions, a comfortable sound environment significantly enhanced TCV. This effect may stem from the psychological benefits of an acoustically pleasant environment, which fosters a more positive mental state and increases tolerance to external conditions. TCV represents a complex perceptual experience that depends not only on physiological factors such as thermoregulation and skin sensation but also on psychological influences. These psychological effects are shaped by multisensory stimuli, individual preferences, and a range of cognitive mechanisms [51]. Psychological factors, in turn, can affect human behavior and interactions with the surrounding environment [52].

ASV and ACV were primarily influenced by the acoustic environment, but certain sound types were also affected by thermal conditions, in line with previous research [53–55]. Under natural sound exposure, ASV decreased as temperature increased, indicating that higher temperatures made the environment feel noisier. This phenomenon may be attributed to the fact that excessive heat induces irritability, leading to a more negative psychological state toward the surroundings. A similar trend was observed for ACV in natural sound environments, where ACV was highest under neutral or moderate thermal stress, but declined as thermal conditions became either excessively hot or cold. This is consistent with Jin et al.'s findings that both high and low temperatures exacerbate acoustic discomfort [50].

In the transitional season, TCV significantly influenced subjective ASV and ACV. Across artificial, natural, and mechanical sound environments, higher TCV corresponded to higher ASV, suggesting that individuals who felt more thermally comfortable perceived their surroundings as quieter, whereas those experiencing thermal discomfort found their environment noisier. This may be because tolerance to noise decreases as thermal discomfort increases. In natural sound environments, ACV increased with TCV, indicating that individuals who felt thermally comfortable also perceived their sound environment as more pleasant. However, when TCV $\leq$ 0, ACV remained at a low level, implying that individuals who experienced thermal discomfort also perceived greater acoustic discomfort. Chinazzo [56] posited that human sensory perception is inherently multimodal, meaning that environmental perception is influenced by multiple sensory inputs simultaneously, a conclusion that aligns with the findings of this study. However, the correlations between thermal and acoustic perception were noticeably weaker in summer and winter. Under such extreme thermal conditions, perception tends to be dominated by thermal sensation itself, leaving limited room for the acoustic environment to modulate TSV, TCV, ASV, or ACV.

4.2. Combined effects of temperature and sound pressure level

Our study demonstrated that SPL significantly influenced TSV and TCV across all three seasons. In hot environments, TSV increased with rising SPL, whereas in cold environments, TSV decreased as SPL increased. This suggests that loud sounds amplify the perception of heat in summer and cold in autumn and winter. In contrast, Guan [57] found that TSV remained unchanged despite variations in SPL, likely because their experiment was conducted within a near-comfort thermal range. In summer, TCV was lowest when SPL ranged from 60 to 70 dB(A), with an average range of 0.4. In winter, TCV increased as SPL decreased, indicating that excessive SPL reduces thermal comfort, a finding consistent with Yang et al. [58].

UTCI significantly influenced ASV in summer and ACV in both summer and winter. Nagano and Horikoshi [59] reported that thermal-acoustic interactions significantly affect auditory perception, aligning with our results. Specifically, in summer, when heat stress levels were

relatively mild (26.5°C–32.0°C), individuals exhibited increased sensitivity to sound, amplifying auditory discomfort—an observation consistent with Wu et al. [60]. Regardless of the season, ACV exhibited an increasing trend with UTCI, indicating that colder conditions intensified auditory discomfort. Jin et al. [50] reported that rising UTCI increased ACV in winter but decreased it in summer, a result that differs from our findings, likely due to variations in the types of sounds used in the experiments.

The interaction between thermal and acoustic environments significantly influenced OCV across all three seasons. Our findings indicate that in summer and transitional seasons, as SPL increased, the differences in overall comfort across temperature levels diminished. Conversely, in winter, rising SPL led to greater disparities in overall comfort across different temperature levels. In both summer and transitional seasons, OCV remained consistently low at higher SPLs, regardless of temperature variations. In winter, when SPL was low, OCV remained high across all UTCI levels, a pattern consistent with Jin et al. [50].

4.3. Evaluation of equivalent OCV under different conditions

Our findings indicate that thermal, acoustic, and combined thermal-acoustic environments significantly influence OCV, consistent with previous studies [42,61]. In the study by Nagano and Horikoshi [62], OCV decreased as ambient temperature increased within the range of 27°C to 39°C, while within the range of 19°C to 28°C, OCV increased accordingly. Furthermore, when either the thermal or acoustic environment reached extreme conditions, OCV remained in an uncomfortable state regardless of variations in other environmental factors. To process the OCV data, we applied Kriging interpolation, a surrogate modeling technique that interpolates limited data points to simulate results across a global domain [63,64].

Fig. 12 illustrates the equivalent comfort zones for overall comfort throughout the year in a park setting. The contour lines were densest in summer, indicating heightened sensitivity to thermal variations, while the contours were sparsest in winter, suggesting reduced sensitivity in colder environments. The distribution in transitional seasons was moderate compared to summer and winter.

As shown in Fig. 12, seasonal variations in the thermal-acoustic environment had a significant impact on OCV. In summer, when UTCI ranged from 26°C to 44°C and L_{Aeq} ranged from 45 dB(A) to 70 dB(A), OCV varied between −3.0 and 3.0. The highest OCV, classified as "very comfortable," occurred when UTCI was below 28°C and L_{Aeq} was below 53 dB(A). When UTCI was below 30°C, OCV remained at or above "moderate" regardless of SPL variations, whereas when UTCI exceeded 42°C, OCV remained below "moderate" regardless of SPL. Notably, when UTCI exceeded 33°C and L_{Aeq} surpassed the environmental noise threshold of 60 dB(A), discomfort was consistently reported. For UTCI below 38°C, each 1.0°C change corresponded to an OCV variation of 0.5, whereas above 38°C, a 0.7°C change resulted in the same OCV variation, suggesting that SPL had a comparatively less significant effect. When L_{Aeq} ranged between 56 dB(A) and 66 dB(A), OCV fluctuated by 0.5, but for L_{Aeq} levels exceeding 68 dB(A) or below 56 dB(A), overall comfort remained unchanged.

During the transitional season, when UTCI ranged from 24°C to 40°C and L_{Aeq} ranged from 45 dB(A) to 75 dB(A), OCV varied between −3.0 and 3.0. The highest OCV, categorized as "very comfortable," was observed when UTCI was below 25°C and L_{Aeq} was below 50 dB(A). When UTCI was below 27°C, OCV remained at or above "moderate" regardless of SPL variations, whereas for UTCI above 38°C, OCV consistently fell below "moderate" regardless of SPL. If UTCI exceeded 30°C and L_{Aeq} surpassed 60 dB(A), discomfort was reported. For UTCI below 35°C, each 6.0°C variation resulted in an OCV change of 0.5, while above 35°C, each 1.0°C-1.5°C change corresponded to an OCV

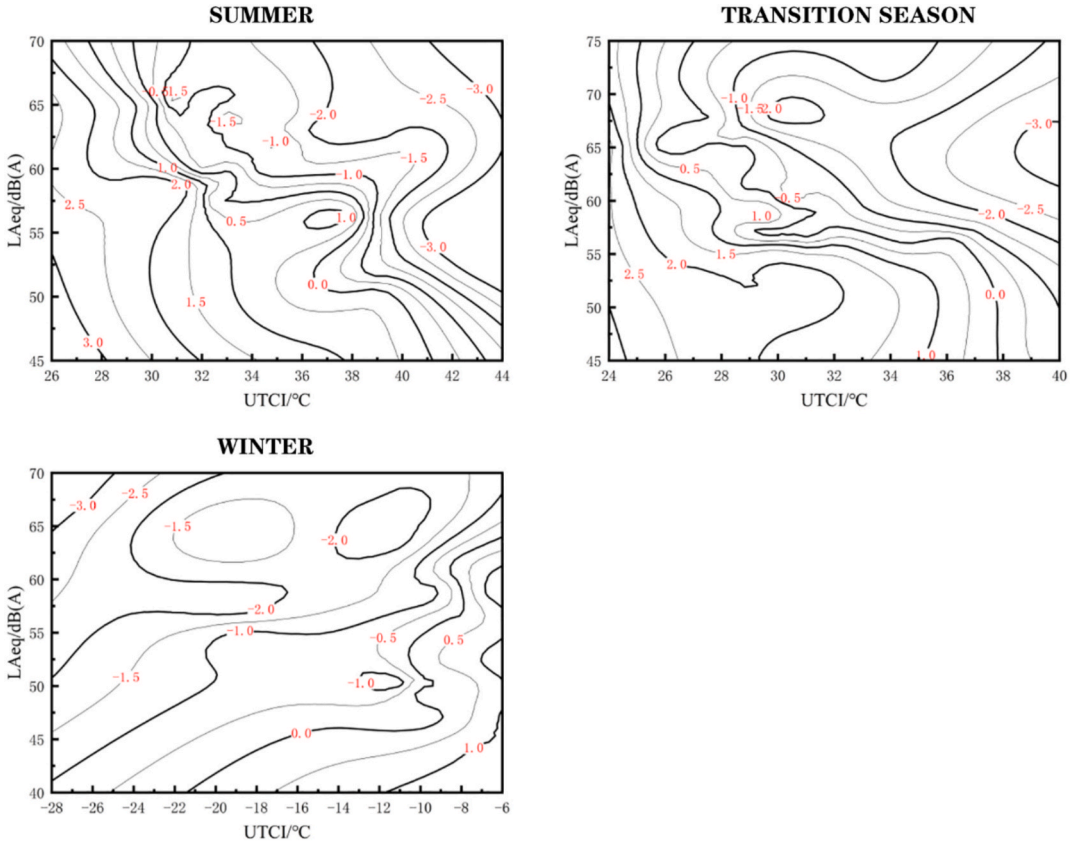


Fig. 12. Equivalent OCV under different conditions of the thermal-acoustic environment.

variation of 0.5. Simultaneously, for every 10 dB(A) change in L_{Aeq}, OCV varied by 0.5.

In winter, when UTCI ranged from −28°C to −6 °C and L_{Aeq} ranged from 40 dB(A) to 70 dB(A), OCV varied between −3.0 and 1.0, with lower values compared to other seasons. When UTCI dropped below –22°C, OCV remained below "moderate" regardless of SPL, and when L_{Aeq} exceeded 63 dB(A), OCV was consistently below "moderate" regardless of temperature variations. Additionally, if UTCI was below −9 °C and L_{Aeq} exceeded 60 dB(A), discomfort was reported. The highest OCV, classified as "slightly comfortable," was observed when UTCI exceeded −12°C and L_{Aeq} remained below 61 dB(A). In winter, each 4.0°C variation in UTCI corresponded to an OCV change of 0.5, while each 6 dB(A) change in L_{Aeq} resulted in an OCV variation of 0.5.

4.4. OCV model

To accurately quantify the relationships between different environmental factors and overall comfort, mathematical models and evaluation formulas can be utilized. In this study, a comprehensive dataset was obtained through field measurements and questionnaire surveys, enabling the construction of an environmental comfort evaluation model using regression analysis. The model was developed to assess overall comfort by integrating both objective environmental parameters and subjective perceptions reported by participants.

Evaluation Model Based on Objective Thermal-Acoustic Parameters:

$$\text{Summer}: \ \text{OCV} = 23.051 - 0.339\text{UTCI} - 0.199\text{L}_{\text{Aeq}} (\text{R}^2 = 0.730) \quad (1)$$

$$\textit{Transitionalseason}: OCV = 15.213 - 0.230UTCI - 0.137L_{Aeq} (R^2 = 0.722) \quad (2)$$

$$\text{Winter}: \ \text{OCV} = 4.750 + 0.136\text{UTCI} - 0.066\ \text{L}_{\text{Aeq}} (\text{R}^2 = 0.768) \quad (3)$$

The regression models reflect the strength and direction of the influence of thermal and acoustic environments on OCV under three distinct climatic conditions.

Effect of the thermal environment (UTCI): In both summer and the transitional season, OCV was negatively correlated with UTCI, indicating that rising thermal stress significantly decreased participants' comfort levels. This effect was most pronounced in summer, where an increase of 1 °C in UTCI led to a 0.339-unit reduction in OCV. Conversely, in winter, UTCI exhibited a positive correlation with OCV, suggesting that higher temperatures improved comfort perceptions (with a 1 °C increase resulting in a 0.136-unit rise in OCV), which aligns with people's preference for warmer conditions in cold climates.

Effect of the acoustic environment (L_{Aeq}): Across all three seasons, L_{Aeq} showed a consistent negative correlation with OCV, demonstrating that increased noise levels reduced perceived overall comfort regardless of climate conditions. The magnitude of this effect varied seasonally, with the strongest impact observed in summer (a 1 dB(A) increase in L_{Aeq} resulted in a 0.199-unit decline in OCV) and the weakest in winter (a 0.066-unit decline). Together, these models confirm the co-regulatory effects of thermal and acoustic environments on subjective comfort, with their relative influence varying by season.

To evaluate the robustness and generalization capability of the regression models across different seasons, we performed 10-fold cross-validation for each of the models developed for summer, autumn, and winter datasets. The R^2 values from each fold were recorded to assess predictive performance and stability. The summer model achieved a mean R^2 of 0.711 with a standard deviation of 0.044, indicating stable performance. The autumn model yielded a mean R^2 of 0.706 and a standard deviation of 0.106, suggesting good overall performance with some variability between folds. The winter model produced R^2 values ranging from 0.377 to 0.882, with a mean R^2 of 0.744 and a standard deviation of 0.140, closely aligning with the training R^2 of 0.770. This indicates that the model generalizes well to new data. In conclusion, the models exhibited stable and consistent performance across seasonal datasets, supporting their effectiveness and strong generalization capabilities.

Evaluation Model Based on Subjective Perception of the Thermal-Acoustic Environment:

$$\textit{Summer}: OCV = 0.445 + 0.879TCV + 0.603ACV (R^2 = 0.708) \quad (4)$$

$$\textit{Transitionalseason}: OCV = -0.157 + 0.669TCV + 0.4479ACV (R^2 = 0.728) \quad (5)$$

$$\textit{Winter}: OCV = 0.646 + 0.781TCV + 0.314ACV (R^2 = 0.751) \quad (6)$$

In all three seasons, OCV was significantly and positively correlated with both TCV and ACV. This indicates that participants who reported higher levels of thermal or acoustic comfort also tended to rate the overall environmental comfort more favorably, confirming the direct enhancing effects of thermal and acoustic perceptions on overall environmental satisfaction. Among the two, TCV consistently exhibited a stronger influence on OCV across all seasons, highlighting the dominant role of thermal perception in shaping overall comfort evaluations. This dominance was particularly pronounced in summer, where the regression coefficient for TCV reached 0.879. In contrast, the effect of ACV was relatively weaker, especially in winter (coefficient = 0.314), suggesting that under cold conditions, the influence of the acoustic environment on perceived comfort diminishes as individuals become more sensitive to thermal conditions.

To evaluate the robustness and generalization ability of the regression models across different seasons, we performed 10-fold cross-validation for each seasonal dataset. The R^2 values from each fold were recorded and summarized. The results indicate that all three seasonal models demonstrate good predictive performance and consistency across folds. The average R^2 for the summer model was 0.696 (SD = 0.082), for the autumn model 0.694 (SD = 0.102), and for the winter model 0.735 (SD = 0.104). Despite some variability, the models remained stable across folds, suggesting strong generalization performance.

4.5. Limitations

Although this study systematically examined the impact of thermal-acoustic environments on overall comfort in cold-region urban parks and classified sounds into natural, human-induced, and artificial types, several limitations remain:

First, spatial representativeness is limited. While 798 valid questionnaires were collected across three seasons with balanced gender and age distributions, most data came from Zhangjiakou, with over 90 % of participants being local residents, limiting geographic and cultural diversity.

Second, comfort perception is highly subjective and context-dependent, influenced by activity type, psychological expectations, and social settings. Although objective indicators such as UTCI and L_{Aeq} were introduced and survey periods controlled, external variability could not be fully eliminated.

Third, environmental variables such as lighting and air quality were not included, and the current sound classification is coarse, limiting the ability to capture the complexity of park soundscapes. Moreover, the distribution of different sound types across specific sound pressure levels (e.g., 40–50 dB, 60–70 dB) was not analyzed. Due to the use of overall L_{Aeq} measurements without source recognition systems, quantitative associations between sound types and sound levels could not be established.

Future research will: (1) expand the sample scope to include diverse urban spaces across different climates and cultural contexts to improve model generalizability; (2) incorporate more refined sound classifications and additional environmental parameters (e.g., lighting, air

quality) to enhance modeling accuracy; and (3) apply source recognition, longitudinal tracking, behavioral observation, and psychological assessment to dynamically explore comfort perception and better understand human–environment interactions.

5. Conclusions

In this study, three representative spaces within an urban park in Zhangjiakou, a cold-region city in China, were selected to investigate people's subjective perception of the thermal-acoustic environment. By integrating on-site meteorological measurements with questionnaire surveys, this study explored the combined effects of the thermal-acoustic interactive environment on subjective evaluations in urban parks located in cold regions of China. The key findings are as follows:

1. Thermal Perception: Sound type and sound pressure level had significant effects on TSV and TCV. In hot environments, exposure to natural sounds significantly reduced subjective TSV and improved TCV. High sound pressure levels ($L_{Aeq} \geq 60$ dB(A)) increased TSV in summer but decreased it in winter while reducing TCV. Additionally, the more extreme the thermal environment, the greater the impact of sound pressure level on TSV.
2. Acoustic Perception: The influence of UTCI on ASV was significant only in summer, where ASV increased with UTCI. The effect of UTCI on ACV was significant in both summer and winter, with ACV increasing as UTCI rose. In a natural sound environment, ASV decreased as temperature increased, while both extremely low and high temperatures led to higher acoustic discomfort. A thermally comfortable environment contributed to higher ASV and increased ACV.
3. Overall Comfort: Both thermal and acoustic environments had significant effects on OCV. In summer and transitional seasons, OCV exhibited a negative correlation with both UTCI and sound pressure level. In winter, OCV was positively correlated with UTCI but negatively correlated with sound pressure level.
4. Thermal-Acoustic Effects in Cold-Region Urban Parks: The impact of high-temperature environments (UTCI $\geq$ 35°C) on OCV during summer and transitional seasons was greater than that of low temperatures in winter. Conversely, variations in sound pressure level had a more pronounced effect on OCV in winter than in summer and transitional seasons. Specifically, in summer, when UTCI exceeded 42°C, OCV remained below the "moderate" level regardless of sound level variations. During transitional seasons, when UTCI exceeded 38°C, OCV remained below "moderate" irrespective of park noise levels. In winter, when UTCI fell below –22°C or L_{Aeq} exceeded 63 dB (A), OCV remained below "moderate" regardless of other environmental conditions.

CRediT authorship contribution statement

Xin Wu: Writing – review & editing, Methodology, Conceptualization. **Chaopu Yuan:** Writing – review & editing, Writing – original draft, Methodology, Data curation. **Haoran Xue:** Software, Project administration, Investigation. **Yang Yan:** Writing – review & editing, Methodology.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix X

Questionnaire on Subjective Evaluation of Urban Park Environment

Survey Point:_____Time of Completion:_____

Dear participant,

This is a questionnaire survey regarding the subjective evaluation of urban park environments. We sincerely appreciate your participation and kindly ask you to answer based on your actual experience. Your responses will greatly contribute to our research on urban park environmental quality. This survey is strictly for academic purposes, with no commercial intent, and all responses will be kept confidential. Thank you for your cooperation!

Part I: Basic Information

- **Gender:** [A] Male [B] Female
- **Age:**_____years
- **Height:**_____cm **Weight:**_____kg
- **Clothing:**
- **Upper body:**_____**Lower body:**_____**Footwear:**_____**Other:**_____
- **Current activity status:**
- [A] Sitting [B] Standing [C] Walking [D] Exercising
- **Duration of residence in Zhangjiakou:**
- [A] Tourist [B] Less than 1 year [C]_____years

Part II: Subjective Environmental Perception and Comfort Evaluation

1 Please evaluate your current subjective sensations and comfort levels:

Scale	−4	−3	−2	−1	0	1	2	3	4
Thermal Sensation	Very Cold	Cold	Cool	Slightly Cool	Neutral	Slightly Warm	Warm	Hot	Very Hot
Scale	**−3**		**−2**	**−1**	**0**	**1**	**2**	**3**	
Acoustic Sensation	Very Noisy		Noisy	Slightly Noisy	Neutral	Slightly Quiet	Quiet	Very Quiet	
Thermal Comfort	Very Uncomfortable		Uncomfortable	Slightly Uncomfortable	Neutral	Slightly Comfortable	Comfortable	Very Comfortable	
Acoustic Comfort	Very Uncomfortable		Uncomfortable	Slightly Uncomfortable	Neutral	Slightly Comfortable	Comfortable	Very Comfortable	
Overall Comfort	Very Uncomfortable		Uncomfortable	Slightly Uncomfortable	Neutral	Slightly Comfortable	Comfortable	Very Comfortable	

Data availability

Data will be made available on request.

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